Teach Guide — Session 3

How the Tree Picks Its Questions

.short slides. Full script: teaching-notes.md 8

Prep checklist (night before)

- PHYSICAL props: three grab-bags — 8 same-color; 4+4; 3+3+2 (or magnets + two drawn boxes)
- PART 0 demo cell runs fresh (the machine's gain leaderboard)
- Judge green: `python judge/run.py notebook/s03_notebook_solution.ipynb` → 3/3
- Board space for the score-card table: question | sides | surprise after | score

Run of show (60 min)

Time	Segment	Where
0–5	Hook: two boxes, one question	board/props
5–15	Purity: sorted boxes	slides 5–8
15–25	Grab-bags → surprise number	props + A2
25–35	THE GAME: teams score questions	board + hands
35–45	Titanic gains	A4–A5
45–50	You be the machine (`best_question`)	A6
50–55	Consolidation quiz	slides
55–60	Ship + handout (one slide)	B1–B3

The one rule of this session

Experience before the name. The team game comes FIRST, the words "information gain" AFTER.

- Slide 12 is the game; slide 13 names it. Never swap them.
- Wait for a team to notice the TIE themselves: `weight ≥ 14` and `ear ≥ 8.25` both score 0.95.
- The tie break is boring and mechanical (column order) — say so. Honesty beats drama.

Key lines to say verbatim

- "A good question doesn't just cut the box — it SORTS."
- On the bad question: "One door IS pure — but it holds one animal out of eleven. The score cares about the whole box."
- On Titanic: "Session 1 you VOTED that gender matters. Today the machine put a number on your vote."
- The dark slide: "«Does it hop?» can never be asked — it isn't a column. Features are OUR job."

Misconceptions to catch

They think	You show
Splitting evenly is the goal	The 1-vs-10 split: a perfect side that earns almost nothing (0.14)
Entropy is the machine thinking	It's counting colors. The least magical number in the course
The champion is the purest side	The champion is the biggest surprise-DROP for the whole box
Gain shows causes	Gain measures sorting — one sentence; Session 9 opens fairness

Live-demo protocol

- PART 0 (leaderboard print), A2/A2b (bag numbers + 11-animal box: 1.57), A3 (the machine's leaderboard), A3b (light-box champion), A4–A5 (Titanic 0.97 → `is_female` 0.21), A6 (the loop).
- If a cell breaks: the same numbers are committed in the solution notebook's outputs and in the slide pictures (split_bad, split_champion_tree, leaderboard_animals, titanic_champion).
- Never debug live longer than 60 seconds.

Judge + homework + cliffhanger

- At-home: B1 (three bags' surprises: 0, 1, 0.47 — **don't round**), B2 (light-box champion), B3 (Titanic leaderboard + hand-annotated plot). ~20–30 min.
- Judge command is ON the closing slide: `python judge/run.py s03_notebook.ipynb`
- **Ship:** the annotated Titanic map, `s03_titanic_map.png` — "my Session 1 vote got its number today."
- Cliffhanger: "What if a question is only 'good' on passengers it has SEEN? Next: the exam cheat test."

Within 48h: retro block in the session README.